

Policy Research on State-Run Groceries

<https://gemini.google.com/app/85fcf8dbf0c2be9c>

User prompt: I am looking to do a policy research project. The requirement is that there is enough studies, data, and thus evidence, supporting and opposing the policy to make it interesting. One topic I am interested in is state-run grocery stores as a way to lower the cost of food. Return academic and non academic studies that refer to government provision of goods in general or specifically grocery items. This is also a complicated topic because some plans, such as in NY, the government covers rent and the private company can operate under a contract of lower prices. What kind of studies should be investigated here? Government contracts, accountability measures (and failures), etc. Do you have any clarifying questions before returning information?

Response: Sorry, something went wrong. Please try your request again.

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Response: Evaluating state-run grocery stores and public-private food retail requires pulling from agricultural economics, public administration, and urban planning. Before pulling a comprehensive bibliography of specific papers, here is a high-level framework of the *types* of studies to investigate, followed by a few clarifying questions to narrow our focus.

1. Public-Private Partnerships (PPPs) and Contracting

When governments subsidize rent or operating costs in exchange for price caps (like the New York proposals), the research shifts from retail economics to contract theory.

- **What to investigate:** Look for literature on "outcome-based contracting," "viability gap funding," and "principal-agent problems" in PPPs. You will need to examine how governments enforce service quality and price caps without over-regulating the private operator into bankruptcy.
- **Proxies to study:** Affordable housing mandates (where developers get tax breaks for lower rents) and municipal broadband contracts.

2. State-Owned Enterprises (SOEs) and Government Retail

If a municipality fully owns and operates the store, you must investigate the efficiency of government-run retail versus private supermarkets, which operate on razor-thin 1-2% profit margins.

- **What to investigate:** The core economic question is whether removing the profit motive and exploiting tax-exempt status outweighs the higher labor costs (union/pension obligations) and potential bureaucratic inefficiencies.
- **Proxies to study:** The Defense Commissary Agency (DeCA) is the most successful U.S. model—a network of over 200 military grocery stores generating billions in revenue. You should also look at state-run liquor monopolies and historical municipal utilities.

3. Market Failure and "Food Desert" Interventions

The justification for government intervention usually hinges on market failure—private stores cannot survive in certain low-income neighborhoods due to high theft, low volume, or poor supply chain logistics.

- **What to investigate:** Look at studies evaluating municipal policy supports for healthy food retailers. For example, local governments have increasingly utilized zoning, tax incentives, and direct grants to attract supermarkets to underserved areas (Onufrak et al., 2023). Because this project will likely require robust data analysis, prioritize empirical papers that utilize Nielsen retail scanner data or ACS microdata to measure price elasticity and neighborhood impact.

Explore how different intervention models shift the financial and administrative burden:

To ensure the specific papers and datasets I return are perfectly aligned with your project, I have a few clarifying questions:

1. **Geographic Scope:** Are you focusing strictly on large U.S. municipal proposals (like Chicago and NYC), looking at regional efforts (like recent proposals in Erie, PA), or are you open to international case studies?
2. **Primary Policy Goal:** Is your research primarily evaluating **cost reduction/affordability** across the board, or is it focused on **geographic access** (eradicating food deserts)?
3. **Intervention Type:** Do you want to weigh full public ownership heavily against public-private partnerships, or explore both equally?

References Onufrak, S. J., Moore, L. V., Pierce, S. L., MacGowan, C. A., & Galuska, D. A. (2023). Changes in Policy Supports for Healthy Food Retailers, Farmers Markets, and Breastfeeding Among US Municipalities, 2014–2021: National Survey of Community-Based Policy and Environmental Supports for Healthy Eating and Active Living (CBS-HEAL). *Preventing Chronic Disease*, 20. <https://doi.org/10.5888/pcd20.230018> Cited by: 7

User prompt: Let's say I was interested in building a model where we insert a subsidized grocery store into an area with 3 other stores that differed in brand, price, quality, and variety. Then we insert another grocery store and people choose based on the above mentioned variables + distance, and I have to calculate how it will affect the market. The different levels are - incentivize with tax breaks, pay rent, fully operate. Has this kind of simulation been done before?

Response: Yes. The exact simulation framework you are describing is a well-established and highly regarded methodology in economics. In the field of Empirical Industrial Organization (IO), what you

have just described is known as a **Structural Estimation and Counterfactual Simulation**.

While researchers have used this exact math to simulate supermarket mergers (like Kroger and Albertsons) or the expansion of Walmart Supercenters, applying it specifically to compare tax breaks vs. rent subsidies vs. a state-run grocer is an excellent, novel application for a policy research project.

Here is how the literature breaks down the simulation you want to build, and the terms you should use to find the right papers.

1. The Demand Side: How People Choose

When you model consumers choosing between 4 stores based on price, quality, variety, brand, and distance, you are describing a **Spatial Discrete Choice Model** (often referred to as a Mixed Logit or BLP model, named after Berry, Levinsohn, and Pakes).

- **How it works in the literature:** The model calculates a "Utility" score for every consumer for every store. Travel cost (distance) acts as a penalty. If the subsidized store is closer but has lower variety, the model calculates exactly what percentage of consumers will still make the trip based on their price sensitivity.
- **What to research:** Search for papers on "*spatial discrete choice models in retail*" or "*structural demand estimation for supermarkets*." A classic application of this is Ellickson, Grieco, and Khwaja (2020), who created a spatial model to measure competition between different grocery formats based on consumer location.

2. The Supply Side: The Market Reaction

When you insert the new store, the three existing stores will not just sit still. If the new subsidized store steals their customers, the incumbents will react by lowering their prices, improving their quality, or—if they lose too much money—closing down.

- **How it works in the literature:** This is modeled as a **Spatial Equilibrium** or an **Entry Game**. The model assumes the private stores are playing a game where they try to maximize their profit based on what the other stores are doing.
- **What to research:** Search for "*discrete games of market entry*" or "*endogenous market structure in retail*." Grieco (2010), for example, used an entry game model to simulate how incumbent grocery stores decide to enter or exit a local market when a massive competitor (like a supercenter) is inserted nearby.

3. The Policy Experiment: The Different Levers

Testing the different policy levers (tax breaks vs. rent subsidies vs. state-operated) is what economists call running a **Counterfactual**. You estimate the model on real data, and then you change the parameters to see what *would* happen.

This is where your project gets highly interesting, because you are altering the mathematical objective of the new store:

- **Tax Breaks & Rent Subsidies:** The new store is still a private company trying to maximize profit, but you lower their "Fixed Cost" parameter. Studies evaluating local government

attempts to eradicate "Food Deserts" often use this counterfactual to see if a subsidy is large enough to induce a private grocer to open a store in a low-income area.

- **Fully Operate (State-Run):** This fundamentally changes the math. A state-run store does not maximize profit. It maximizes *consumer welfare* (keeping prices as low as possible) subject to a budget constraint (breaking even). Because the state store prices its goods at-cost, it forces the three private incumbents to slash their prices much more aggressively than they would if a normal private competitor had entered.

The "Food Desert" Warning

One major finding in the literature you must account for in your simulation is **Tiebout Sorting** and consumer preference constraints.

Several studies have shown that when local governments heavily subsidize a new grocery store in an underserved neighborhood, consumers often *continue* to travel further away to their old grocery stores. Even though the new store is closer (better distance variable), the consumers had established preferences for the brand, specific ethnic varieties, or higher quality of the distant store. Your model must allow for strong brand loyalty, otherwise, it will over-predict how successful the subsidized store will be.

Sources

1. [Measuring competition in spatial retail - Penn State](#)

User prompt: Could I simulate a simple version of this using gurobi optimization or would I need a more advanced simulation software and advanced degree in economics?

Response: You do not need an advanced degree in economics or specialized software like Stata/Matlab to build a compelling simulation. You can build a stylized, policy-relevant version of this model in Python using Gurobi or standard optimization libraries (`scipy.optimize`, `Pyomo`).

In academic economics, full structural models (like BLP) are built to estimate unknown parameters from raw historical data. For a policy research project, you can flip the workflow: **assume reasonable baseline parameters** for consumer preferences and store costs, and run a simulation to evaluate trade-offs across different policy levers.

Two Ways to Build It in Python

Approach 1: The Deterministic Facility Location Model (Pure Gurobi MIP)

If you want to use Gurobi directly, you can frame this as a spatial facility location and assignment problem with mixed-integer programming (MIP).

- **Consumer Choice:** Divide your geographic area into grid points (or census block groups) i . Each grid point assigns its population to the single store j that offers the highest linear utility score:

$$U_{ij} = \alpha \cdot \text{Quality}_j + \beta \cdot \text{Variety}_j - \gamma \cdot \text{Price}_j - \delta \cdot \text{Distance}_{ij}$$

- **Market Equilibrium:**

1. Solve for baseline consumer utility and incumbent profits across the 3 private stores.
2. Insert Store 4 under three distinct mathematical formulations:

- **Tax Break Scenario:** Store 4's fixed cost F_4 is reduced by T . Store 4 sets price P_4 to maximize profit $\pi_4 = (P_4 - c_4) \cdot Q_4 - (F_4 - T)$.
- **Rent Subsidy Scenario:** Government covers rent R . Store 4 operates as a private firm under a contracted maximum price cap $\bar{P}_4$.
- **State-Run Scenario:** Store 4's objective function switches from profit maximization to **Consumer Welfare Maximization** ($\sum_i Q_{i4} U_{i4}$) subject to a zero-profit or loss-cap constraint ($\pi_4 \geq -\text{Subsidy}$).

- **Why Gurobi excels here:** Gurobi handles binary store-choice variables ($x_{ij} \in \{0, 1\}$) and piecewise linear approximations of price responses at scale.

Approach 2: The Logit Demand + Nash Equilibrium Micro-Simulation (Python + SciPy)

If you want smooth, probabilistic consumer choices where people split their shopping trips across multiple stores, a Multinomial Logit (MNL) framework in Python is often easier than Gurobi.

1. **Probability Formulation:** The probability P_{ij} that a consumer at location i chooses store j is:

$$P_{ij} = \frac{\exp(U_{ij})}{\sum_{k=1}^K \exp(U_{ik})}$$

2. **Iterative Price Best-Response (Nash Equilibrium):**

- Calculate market share $Q_j = \sum_i \text{Pop}_i \cdot P_{ij}$.
- Private stores set prices where marginal revenue equals marginal cost:

$$P_j = c_j + \frac{1}{\gamma(1 - S_j)}$$

- Run a simple `while` loop in Python: update Store 4's policy, let private stores re-optimize their prices in response, and iterate until prices stabilize (convergence to Nash equilibrium).

Comparing the Implementation Paths

Feature	Gurobi (Deterministic MIP)	Python/SciPy (Logit Micro-Simulation)
Consumer Choice	"All-or-nothing" to the single best store per location.	Probabilistic split across multiple stores based on distance and preferences.
Core Strength	Optimizes exact policy bounds (e.g., <i>Find the minimum subsidy S needed to lower market prices by 10%</i>).	Captures smooth shifts in market share and competitive price responses.
Setup Complexity	Moderate (defining linear/MIP constraints).	Low to Moderate (uses standard matrix operations in numpy / scipy).

What Makes This Rigorous for Policy Analysis

You do not need real structural econometric estimation to make this project valuable. The strength of your project comes from **sensitivity analysis**:

- How high does a tax break need to be before a private grocer lowers prices as much as a state-run grocer would?
- At what distance penalty δ do consumers in low-income zones stop switching to the subsidized store?
- How much profit do incumbent private stores lose before one of them is forced to exit the market entirely?

User prompt: Here is another policy I would like to research (not simulate). security policy in mexico. The policy question is: what policy has been implemented in Mexico in 2026 to reduce crime, and what are the arguments for/against, and what alternatives should be pursued

Response: Under President Claudia Sheinbaum, Mexico’s 2026 security framework has explicitly pivoted away from her predecessor’s passive approach toward a model of constitutional militarization combined with intelligence-led, regionalized offensives.

1. The 2026 Policy Implementation

The current strategy rests on two primary pillars implemented through a wave of secondary legislation passed in mid-2026:

- **Complete Militarization of the National Guard:** In July 2026, Congress passed secondary legislation finalizing a constitutional amendment that formally transferred the National Guard to the Secretariat of National Defense (SEDENA). It is now legally the fourth branch of the armed forces. This legislation grants the military sweeping new civilian law enforcement

powers, including authorization to intercept private communications and conduct undercover operations.

- **Civilian Intelligence and Regional Offensives:** Led by Security Minister Omar García Harfuch, the government passed the Law on the National Intelligence and Investigation System to build a specialized, civilian-led investigative unit. Operationally, this translates into "Unified Commands" and regionalized surges, such as the comprehensive *Michoacán Plan for Peace and Justice*, which coordinated state police and military assets to simultaneously target 12 specific cartels while deploying billions of pesos in infrastructure and social programs.

2. The Debate: Arguments For and Against

Arguments Supporting the Policy:

- **Operational Efficiency & Metric Reductions:** Proponents argue that bringing the National Guard under military command eliminates bureaucratic friction. Early localized data supports this: in July 2026, the State of Mexico reported a 31% year-over-year drop in daily homicides after implementing a unified command structure in high-crime municipalities.
- **Increased Seizures and Arrests:** Sheinbaum's administration has aggressively utilized military deployments to combat organized crime, resulting in record levels of drug seizures (over 245 tonnes) and the detention of tens of thousands of suspects tied to criminal organizations.

Arguments Opposing the Policy:

- **Erosion of Civilian Oversight:** Critics argue the July 2026 secondary laws distort public security by establishing a paradigm where soldiers investigate civilians. The simultaneous dissolution of the autonomous transparency institute (INAI) severely limits the ability of civil society to monitor military abuses or human rights violations, which remain widespread.
- **The Judicial Reform Paradox:** The strategy relies heavily on mass arrests, yet the justice system lacks the capacity to prosecute them. Prosecutors currently solve only about one in ten intentional homicides. Furthermore, the constitutional mandate forcing all federal judges to stand for popular election (with the first round held in June 2026) has severely destabilized the judiciary, raising deep concerns about institutional integrity, judicial independence, and vulnerability to cartel influence.
- **Targeting Foot Soldiers over Networks:** Despite thousands of arrests, the operations primarily capture low-level operatives for weapons or drug possession rather than dismantling the financial and political networks of high-ranking criminal figures.

3. Alternatives for Policy Research

A robust policy analysis of the 2026 landscape should evaluate the following counterfactuals and alternative approaches:

- **Rebuilding Local and State Police:** The federalization of security has historically starved municipal and state police of funding. An alternative policy model would redirect federal

military budgets into localized capacity building, ensuring that first responders understand the community dynamics and are accountable to local jurisdictions rather than federal generals.

- **Targeting Illicit Financial Flows:** Rather than kinetic military operations that fracture cartels and increase localized violence, an alternative strategy would prioritize the Financial Intelligence Unit (UIF). This means shifting resources toward forensic accounting, money laundering prosecutions, and seizing cartel assets, which neutralizes organizational capacity without street-level combat.
- **Prosecutorial Capacity Building:** The bottleneck in Mexican security is not a lack of arrests, but a lack of specialized investigators at the *Ministerio Público* level. Policy alternatives should explore shifting investment from the National Guard into training and protecting prosecutors, creating independent accountability mechanisms, and utilizing data-driven triage to prioritize complex organized crime cases over petty offenses.

Sources

1. [Mexico's new security laws expand military power and erode civilian](#)
2. [Security Reforms - Mexico Business News](#)
3. [Sheinbaum announces 'Plan for Peace and Justice' - EFE](#)
4. [México state security strategy has slashed homicides in half](#)
5. [Mexico: Backing Security Reform as U.S. Pressure Mounts](#)
6. [World Report 2026: Mexico | Human Rights Watch](#)

User prompt: What international evidence supports or refutes this strategy? Provide links to sources

Response: 1. Evidence Refuting the Militarized Policing Model

Colombia's "Plan Fortaleza" (Null Crime Reduction) A rigorous study published in *Nature Human Behaviour* evaluated a massive military policing intervention in Cali, Colombia, a city with historically high violent crime rates. Researchers from Brown University and Universidad de los Andes found scant evidence that intensive military patrols reduced the prevalence of crime. In fact, crime slightly increased after the intervention ended, suggesting that militarization acts merely as temporary displacement rather than a structural solution.

- [Researchers find little evidence that military policing reduces crime \(Brown University\)](#)

Broader Latin American Trends (Human Rights and Democratic Erosion) A comprehensive analysis published in *Perspectives on Politics* evaluates the "constabularization of the military" across Latin America, including Brazil's use of troops in Rio de Janeiro and Mexico's ongoing security strategies. The study concludes that using military forces for domestic law enforcement consistently undermines citizen security, stalls police reform, and erodes the legal order. Furthermore, a report by the Washington Office on Latin America (WOLA) explicitly warns that militarization without strict civilian oversight directly correlates with severe human rights abuses, including torture and extrajudicial executions, while leaving the underlying rate of impunity unchanged.

- [The Militarization of Law Enforcement: Evidence from Latin America \(Cambridge University Press\)](#)
- [Militarization Without Civilian Controls Puts Human Rights at Risk in Mexico \(WOLA\)](#)

2. Evidence Supporting the Militarized/Heavy-Handed Model

El Salvador's "Plan Control Territorial" (Rapid Homicide Reduction) The most potent international evidence supporting a heavily militarized, mass-arrest strategy comes from Nayib Bukele's administration in El Salvador. By utilizing the military for domestic policing, imposing a prolonged state of exception, and conducting mass detentions of suspected gang members, El Salvador's homicide rate plummeted from 38 per 100,000 in 2019 to just 2.4 per 100,000 in 2023.

- **The Trade-off:** Researchers note this unprecedented drop relies entirely on physical incapacitation (removing tens of thousands of young men from the streets) and centralized political control. It has come at the cost of sweeping human rights violations, the suspension of due process, and a ballooning prison population that poses severe long-term sustainability risks.
- [Why 'Plan Bukele' Does \(and Doesn't\) Work \(Global Americans\)](#)

3. Evidence Supporting the Policy Alternatives

Targeting Illicit Financial Flows (IFFs) Over Kinetic Combat International bodies like the United Nations Office on Drugs and Crime (UNODC) and the Financial Action Task Force (FATF) provide extensive evidence that targeting the financial infrastructure of organized crime is far more effective at dismantling cartels than street-level military combat. Kinetic military operations in Mexico have historically triggered the "kingpin effect"—where capturing or killing a cartel boss fractures the group, leading to violent succession wars and increased civilian casualties. UNODC frameworks demonstrate that starving organizations of Illicit Financial Flows disables their capacity to purchase weapons, bribe officials, and maintain logistics networks without sparking retaliatory street-level violence.

- [Illicit Financial Flows Frameworks \(UNODC\)](#)

Civilian Investigative Capacity (The CICIG Model) While highly context-specific, the International Commission Against Impunity in Guatemala (CICIG) remains a primary benchmark for the alternative of building civilian prosecutorial capacity. By embedding international investigators directly with local prosecutors, CICIG helped dismantle complex criminal networks from the top down and contributed to a significant, sustained reduction in Guatemala's homicide rate by strengthening the rule of law rather than relying on military force.

Sources

1. [Researchers find little evidence that military policing reduces crime](#)
2. [The Militarization of Law Enforcement: Evidence from Latin America](#)
3. [Militarization Without Civilian Controls Puts Human Rights at Risk in](#)
4. [Why 'Plan Bukele' Does \(and Doesn't\) Work - Global Americans](#)

User prompt: Is there any mexican policy in particular that deals with business extortion? or is it mostly just towards overall security?

Response: Yes, the Sheinbaum administration has recently carved out business extortion from the broader security umbrella, treating it as a specialized economic and social crisis.

Historically, extortion—specifically *cobro de piso* (charging businesses a "protection tax")—was treated as a local state-level crime. However, the federal government launched the **National Strategy Against Extortion** in July 2025, followed by the enactment of the **General Law Against Extortion** in November 2025.

This new framework introduces several major policy mechanisms specifically designed to protect businesses:

1. Ex Officio Prosecution

Previously, investigating extortion required the victim to file a formal complaint—a massive bottleneck, as business owners understandably feared cartel retaliation. Under the new General Law, extortion must be prosecuted *ex officio* across all 32 states.

When analyzing business owner cases from news media, this legal shift is a critical variable. It means authorities are now legally required to investigate extortion based on public knowledge—such as local news reports or social media videos—without waiting for a terrified business owner to walk into a police station.

2. Explicit Recognition of "Cobro de Piso"

The 2025 legislation specifically codifies business extortion tactics as aggravating circumstances that carry up to 25 years in prison. The law explicitly targets:

- Standard *derecho de piso* (protection fee) schemes.
- Monopoly extortion (forcing a business to exclusively purchase goods, like avocados, tortillas, or lime, from a cartel-approved supplier).

3. Federalization and Pretrial Detention

The reform elevated extortion to a federal crime and added it to the list of offenses requiring mandatory pretrial detention. By federalizing the crime, the government acknowledged that municipal and state police forces are often too compromised or outgunned by local criminal groups to effectively protect small businesses.

4. Disrupting the Financial Logistics

The strategy recognizes that business extortion relies on a massive logistical network to collect and launder cash. Security Minister Omar García Harfuch's strategy includes collaborating with the private sector to train cashiers at convenience stores (like Oxxo) and banks to identify and halt suspicious extortion deposits. The government is also conducting specialized raids on federal and

state prisons, as a vast amount of "virtual extortion" (phone threats to businesses) originates from inmates using contraband cell phones.

Sources

1. [How to stop extortion in Mexico](#)
 2. [Mexico: New General Anti-Extortion Law | Insight | Baker McKenzie](#)
 3. [Mexico launches national strategy against extortion](#)
-

User prompt: What specific policies were enacted in 2025-2026 in Mexico that changed security?

Response: President Claudia Sheinbaum's administration enacted several sweeping legislative packages across 2025 and 2026 that fundamentally altered civil-military relations, intelligence gathering, and public security in Mexico.

National Guard Militarization (July 2026)

Following a 2024 constitutional amendment, Sheinbaum pushed through secondary legislation in July 2026 that formally transferred the civilian-created National Guard to the [Secretariat of National Defense](#).

- **Military Status:** The National Guard is now officially established as Mexico's fourth armed force, operating alongside the Army, Navy, and Air Force.
- **Expanded Powers:** The reform authorizes the Guard to conduct civilian investigations, lead undercover operations, and intercept private communications without requiring prior judicial approval.
- **Jurisdiction:** The reform extends military jurisdiction to National Guard members, meaning they operate under military discipline rather than civilian accountability structures.

The National Intelligence and Investigation System Law (2025–2026)

Referred to by critics as the "Ley Espía" (Spy Law), this legislation constructs a centralized data-sharing framework under the [Security and Citizen Protection Ministry](#).

- **Central Intelligence Platform:** It mandates the integration of data from all levels of government and private entities into a single platform utilizing AI processing.
- **Warrantless Access:** Authorities can access personal, fiscal, biometric, and geolocation data from public institutions and private companies (such as banks and telecom providers) without a prior court order.
- **Mandatory Updates:** Private and public entities are required to update this information daily.

Biometric and Identity Integration (Mid-2026)

To complement the new intelligence frameworks, the administration enacted amendments to the [General Population Law and the General Law on Forced Disappearances](#).

- **Biometric CURP:** The reform mandates the integration of biometric data (fingerprints or facial recognition) with Mexico's national identification number (CURP).
- **Single Identity Platform:** This creates real-time cross-referencing capabilities across the National Registry of Missing People, forensic databases, and civil registries.

The General Law Against Extortion (November 2025)

To address the logistical and financial control cartels exert over local economies, this law structurally changed how the state prosecutes the *cobro de piso* (protection tax).

- **Federalization:** The [Senate advanced legislation](#) establishing extortion as a federal crime, which triggers mandatory pretrial detention for suspects.
- **Ex Officio Prosecution:** The state must now investigate extortion cases automatically upon becoming aware of them, removing the burden from victims who previously had to file formal complaints.

Sources

1. [Mexico's new security laws expand military power and erode civilian](#)
2. [These are Mexico's five new public security laws, and this is what](#)
3. [How Mexico's New Intelligence Law Will Impact Public Security](#)

User prompt: Can you explain what all the variables mean in the problem formulation you provided?

Response: Here is the breakdown of the variables from the simulation models, separated into how consumers make choices (Demand) and how the stores and government operate (Supply & Policy).

1. Indices (The Setup)

- i : A specific consumer location (e.g., a census block group or neighborhood grid point).
- j : A specific grocery store (Stores 1, 2, and 3 are private incumbents; Store 4 is the subsidized/state-run store).

2. The Demand Side (Consumer Choice)

The equation $U_{ij} = \alpha \cdot \text{Quality}_j + \beta \cdot \text{Variety}_j - \gamma \cdot \text{Price}_j - \delta \cdot \text{Distance}_{ij}$ calculates the "utility" (overall appeal) of store j for a consumer at location i .

- U_{ij} (**Utility**): The final score. Consumers want to maximize this number. If using the Gurobi MIP model, all consumers at location i simply go to the store with the highest U_{ij} .
- $Quality_j, Variety_j, Price_j$: The observable characteristics of each store.
- $Distance_{ij}$: The travel distance or time between neighborhood i and store j .
- $\alpha, \beta, \gamma, \delta$ (**The Parameters**): These are the most critical numbers in your simulation. They represent how much consumers *care* about each trait.
 - γ (Price Sensitivity): If this is high, consumers will abandon their favorite store for a cheaper one.
 - δ (Distance Penalty): If this is high, consumers refuse to travel far. A high δ is what creates a "food desert" mathematically, because people are trapped shopping at whatever is closest.

3. The Supply Side (Store Operations)

The equation $\pi = (P - c) \cdot Q - F$ calculates how much money a store makes.

- π_j (**Profit**): The total profit for store j . Private stores make pricing decisions to maximize this number.
- P_j (**Price**): The retail price the store charges consumers.
- c_j (**Marginal Cost**): The wholesale cost the store pays to acquire the food.
- Q_j (**Quantity/Demand**): The total number of customers (or volume of groceries) the store secures based on its market share.
- F_j (**Fixed Cost**): The cost to keep the doors open regardless of how much food is sold (rent, electricity, basic labor).

4. The Policy Levers (Your Interventions)

These are the variables you manipulate for Store 4 to test your different policy interventions:

- T (**Tax Break**): A reduction in the fixed cost ($F_4 - T$). Store 4 still tries to maximize profit, but they need fewer customers to break even, allowing them to survive in a lower-income area.
- R (**Rent Subsidy**): Similar to a tax break, the government covers a specific portion of F_4 .
- $\bar{P}_4$ (**Price Cap**): A maximum price limit set by a government contract in exchange for the rent subsidy R . The store maximizes profit, but is constrained by $P_4 \leq \bar{P}_4$.

- **Zero-Profit Constraint ($\pi_4 \geq 0$):** Used in the state-run model. Instead of maximizing profit, the state-run store acts to maximize consumer utility U_{ij} (by keeping prices as low as possible) while ensuring its revenue exactly covers its operating costs.

User prompt: I would like to make a new model. My idea is to create a model for workforce development nonprofit deciding how to best optimize their offerings for training programs. Each person can be modeled as having different components of human capital. Some skills are transferrable and others are not. It costs money to start a new training program, but may cost less to partner with an existing employer and offer an internship. Some industries have high start up costs but also high gains. some industries might be growing while others have a chance of disappearing. AI adds uncertainty. Which programs are high risk and high reward? which ones are more stable but may not lead to higher earnings? Can an economic model be created in order to optimize for different levels of risk/reward profiles?

Response: Yes, this can be formulated as a **Stochastic Multi-Objective Portfolio Model**—essentially adapting Markowitz Portfolio Theory and Human Capital Theory (Becker) to a non-profit budget constraint.

Instead of maximizing financial ROI for a stock portfolio, the non-profit maximizes **expected net wage gains (social return)** while managing a **variance constraint (risk of industry decay or AI displacement)**.

1. Categorizing the Programs: Risk vs. Reward Spectrum

When modeling workforce development, programs fall along a spectrum defined by capital intensity, skill transferability, and exposure to technological disruption.

Risk/Reward Profile	Program Examples	Economic & AI Characteristics	Cost Structure
High Risk / High Reward	Junior Software Engineering, Cybersecurity, AI Workflow/Prompt Specialist	High Wage Premium, High Disruption: High starting wages, but elevated risk of AI automating junior-level tasks or rapid skill decay.	High F_k (In-House): Requires specialized labs, software, and high-salaried instructors.
Moderate Risk / High Growth	Advanced Manufacturing Technicians, Clean Energy / EV Maintenance	High Transition Gain: High demand driven by federal/industrial investment, but exposed to regional economic shifts.	Mixed: High equipment costs if built in-house; low cost if structured as employer apprenticeships.

Risk/Reward Profile	Program Examples	Economic & AI Characteristics	Cost Structure
Stable / Moderate Reward	Skilled Trades (HVAC, Plumbing, Electrician), Healthcare (LPN, Dental Hygienist, Phlebotomy)	Low Disruption, High Transferability: High physical dexterity and licensing barriers shield them from AI. Steady demand and low variance in placement.	Moderate F_k: Can often leverage community college infrastructure or union apprenticeships.
Stable / Lower Yield	Administrative Support, Logistics/Warehouse Ops, Customer Care	High Automation Risk, Low Barrier: Quick to train and place, but low wage growth and high vulnerability to software/AI agent replacement.	Low F_k / Low c_k: Low setup costs; easily scalable via employer partnerships.

2. Mathematical Problem Formulation

Sets and Indices

- $i \in \{1, \dots, N\}$: Trainees.
- $k \in \{1, \dots, K\}$: Potential training programs.
- $\omega \in \Omega$: Future states of the economy (capturing AI disruption severity and macroeconomic growth).
- $p(\omega)$: Probability of state ω occurring ($\sum_{\omega} p(\omega) = 1$).

Parameters

- **Human Capital:** Individual i has an initial skill vector $\mathbf{h}_i = \begin{bmatrix} h_i^{\text{trans}} \\ h_i^{\text{spec}} \end{bmatrix}$, where h_i^{trans} represents general transferable skills (literacy, numeracy, soft skills) and h_i^{spec} represents industry-specific skills.
- **Skill Gain:** Program k transforms skills by $\Delta \mathbf{h}_k = \begin{bmatrix} \Delta h_k^{\text{trans}} \\ \Delta h_k^{\text{spec}} \end{bmatrix}$.
- **Program Costs:**
 - F_k : Fixed setup cost to launch program k . ($F_k^{\text{In-House}} \gg F_k^{\text{Partner}}$).
 - c_k : Variable cost per trainee assigned to program k .

- **Economic Outcomes:**

- $W_k(\mathbf{h}, \omega)$: Expected wage outcome for a worker with skill vector $\mathbf{h}$ in program k under state ω .
- $\phi_k(\omega) \in [0, 1]$: Employment probability (reflecting AI displacement or industry contraction in state ω).

- **Budget & Capacity:**

- B : Total non-profit budget available.
- Cap_k : Maximum capacity of program k .

Decision Variables

- $x_k \in \{0, 1\}$: Binary variable indicating whether program k is funded/launched.
- $y_{ik} \in \{0, 1\}$: Binary variable indicating whether individual i is assigned to program k .

The Optimization Framework

1. Individual Net Value Function

The expected wage gain for trainee i in program k under state ω is:

$$\Delta V_{ik}(\omega) = \phi_k(\omega) \cdot W_k(\mathbf{h}_i + \Delta \mathbf{h}_k, \omega) - W_{\text{base}}(\mathbf{h}_i, \omega)$$

2. Expected Social Return (μ)

The expected total wage increase across all trainees and states of the world:

$$\mathbb{E}[R] = \sum_{\omega \in \Omega} p(\omega) \left(\sum_{i=1}^N \sum_{k=1}^K y_{ik} \cdot \Delta V_{ik}(\omega) \right)$$

3. Portfolio Variance / Risk (σ^2)

The variance in total outcomes across economic states captures technological and macroeconomic uncertainty:

$$\text{Var}(R) = \sum_{\omega \in \Omega} p(\omega) \left(\left(\sum_{i=1}^N \sum_{k=1}^K y_{ik} \cdot \Delta V_{ik}(\omega) \right) - \mathbb{E}[R] \right)^2$$

Objective Function: Mean-Variance Optimization

The non-profit optimizes a weighted combination of expected wage gains and risk tolerance governed by a risk-aversion parameter $\lambda \geq 0$:

$$\max_{\{x_k, y_{ik}\}} \mathbb{E}[R] - \lambda \cdot \text{Var}(R)$$

Subject to constraints:

1. **Budget Constraint:**

$$\sum_{k=1}^K \left(F_k x_k + c_k \sum_{i=1}^N y_{ik} \right) \leq B$$

2. **Program Launch & Capacity Linkage:**

$$\sum_{i=1}^N y_{ik} \leq \text{Cap}_k \cdot x_k \quad \forall k$$

3. **Single Assignment per Trainee:**

$$\sum_{k=1}^K y_{ik} \leq 1 \quad \forall i$$

4. **Minimum Placement Floor (Baseline Impact Guarantee):**

$$\sum_{i=1}^N \sum_{k=1}^K y_{ik} \cdot \phi_k(\omega) \geq M_{\text{target}} \quad \forall \omega$$

3. Key Insights Provided by the Model

1. **The Efficiency Frontier (λ Sweep):** By varying λ from 0 (risk-neutral) to a high positive number (risk-averse), the non-profit traces out a **Pareto Frontier**. This shows the exact wage trade-off required to protect trainees from AI disruption.
2. **Optimal Hedges:** The model naturally favors bundling **high-risk/high-reward tech programs** with **low-risk/high-stability skilled trades or healthcare roles** to balance overall portfolio variance.
3. **Employer Partnerships as Risk Transfers:** Employer-partnered internships lower fixed costs (F_k) and shift the demand risk onto the employer, allowing the non-profit to allocate remaining funds toward higher-yielding programs.

User prompt: I would like to export this text and equations into a google doc. how to do?

Response: I have created the Google Doc containing the text and equations for the Workforce Development Optimization Model for you.

[Workforce Development Optimization Model](#)

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User prompt: With regard to the grocery problem, here is what I'm thinking to use to help create the model. What data sets exist that I can use, and what other components should I consider (or not consider) in the model? Samuel's ideas: Problem Definition Define a region in NYC: collect demographic data (income, age, average rent, population size) Identify the grocery stores in the region, defined as selling enough food to sustain (produce, meat, dairy, grocery) Square footage of store Type of store: national chain, regional chain, locally owned Using publicly available data Grocery margins Public national chain financial data Grocery industry estimates Elasticity of demand of groceries Estimate the firm profit function using the above variables Assume Not all products have to be modeled individually. A store can have a price, quality, and variety index. Some people prefer to pay more for quality groceries Some people would rather travel far to save money on groceries Companies want to maximize profits Private companies will operate more efficiently than the publicly run store Customers want to optimize their utility Question: consumer preferences are heterogeneous - might have to play around with these parameters to see how they affect the end result. E.g. if consumers are more price conscious, the subsidies may make sense. If consumers are more brand/quality conscious, the spending may not be worth it. National chains have lower cost of goods due to buying in bulk Policy levers: Select which store gets the subsidy (national chain or locally owned) Adjust how much money is spent on the store: Tax break, rent subsidy, full operation Outcomes Consumer welfare Company profits Store closures (can private companies compete with the subsidized prices?)

Response: Datasets for the NYC Grocery Model

- **Store Locations & Characteristics:** The **State of New York Open Data** portal hosts the "Retail Food Stores" dataset, which lists every retail food store licensed by the NYS Department of Agriculture and Markets. This dataset includes operation type, exact coordinates, and crucial capacity metrics like square footage.
- **Demographics & Spatial Access:** The **USDA ERS Food Access Research Atlas** provides census-tract level metrics on distance to supermarkets, crossing spatial data with household resources like family income, poverty rates, and vehicle availability. You can combine this with census block data from the American Community Survey (ACS) to capture neighborhood-specific rent and population density.
- **Pricing, Demand & Elasticity:** The gold standard for estimating the profit function and demand elasticity is the **NielsenIQ Retail Scanner Data**, which tracks weekly pricing and volume from retail point-of-sale systems. Nielsen also offers the **Homescan Consumer Panel**, which tracks individual household purchase histories over time. This panel data allows you to measure exactly what you hypothesized: how heterogeneous consumers react to price versus quality and distance. Access to Nielsen data typically requires a university subscription through the Kilts Center for Marketing.

Components to Add to the Model

- **Capacity Constraints (Congestion):** If you introduce a heavily subsidized, low-price store, consumer utility maximization will drive massive demand to that location. If you do not constrain store capacity (using the square footage variable), your model will unrealistically predict that an impossible volume of people will shop at the single subsidized store. You must include a penalty for store crowding.
- **SNAP/EBT Market Share:** In lower-income NYC census tracts, a significant percentage of grocery revenue comes from SNAP benefits. Your model should segment consumers based on SNAP utilization, as these shoppers have high price sensitivity but zero flexibility to shop at stores that do not accept EBT.
- **Neighborhood-Level Commercial Rent:** Fixed costs in NYC vary drastically by borough and neighborhood. A rent subsidy in one census tract requires exponentially more capital than one a few miles away. You must incorporate a local commercial rent index to accurately evaluate the financial burden of the "Rent Subsidy" policy lever.

Components to Exclude or Simplify

- **Upstream Supply Chain:** Exclude the logistics of warehousing and trucking. Your assumption that national chains have lower costs due to bulk buying should simply be represented as a statically lower marginal cost parameter, rather than modeling the distribution network itself.
 - **Individual Product Granularity:** Stick strictly to your assumption of a price/quality/variety index. Compress the inventory into a single "Representative Basket Price."
 - **Dynamic Market Exit:** For the initial model, assume the incumbent stores are fixed in their locations and only adjust their prices in response to the new store. Modeling immediate store closures introduces a complex dynamic game. Instead, find the price equilibrium first, and evaluate store closure risk post-hoc by checking if incumbent profits drop below zero.
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